



Пирамида данных в робототехнике

О узком месте данных в робототехнике и типах данных, которые объединяются для его решения

ТАНАЙ ДЖАЙПУРИЯ
28 ИЮЛЯ 2026 Г.

🔊 Посл

Я Танай Джайпурия, партнер [Wing](#), и это еженедельный информационный бюллетень о бизнесе технологической индустрии. Чтобы получать рассылку от Тапау на свой почтовый ящик, подпиш. здесь бесплатно:

Введите свой адрес электронно

Подписаться

Привет, друзья!

Большие языковые модели Frontier были предварительно обучены на триллионах текстовых фрагментов, большая часть которых, по крайней мере на начальном этапе, была взята из интернета. По сути, интернет бесплатным корпусом для предварительного обучения, который человечество собрало коллективно.

В робототехнике такого эквивалента нет. Как правило, это означает, что все попытки решить проблему и создать физический СИИ в той или форме связаны с преодолением этого «узкого места» в данных. В этой статье я подробнее расскажу об этом узком месте и представлю свою версию пирамиды данных для роботов, в которой показаны различные виды данных, используемых в обучении роботов, которые можно объединить для решения этой проблемы.



РОБОТОТЕКА

[Open X-Embodiment](#), крупнейший открытый набор данных о роботах, содержит около 1 млн траекторий для 22 типов роботов. Он был собран

исследова
десятков л
медленно

Сергей Лев
рассказал,
для обуче

«Возмож
использ
в прошл
составл

И это даже
содержат
которых з
меньше и

По имеющ

роботах, а для того, чтобы приблизиться к показателям Левина на од
два порядка, потребуется около 1 миллиона часов данных о роботах.
достичь масштабов интернета, похоже, практически невозможно.
Например, на YouTube ежегодно добавляется около 300 миллионов ча
видео. Не существует такой активности, которая позволила бы
пользователям естественным образом собрать корпус данных о робо
как это происходит в интернете с большими языковыми моделями,
поэтому специалисты в этой области признают, что нужны другие
подходы.



Узнайте больше из рассылки от Tanay

Размышления о технологиях и бизнесе

Более 12 000 подписчиков

Введите свой адрес электронной почты...

Подписаться

или



Подписаться с помощью Substack

Подписавшись, вы соглашаетесь с [Условиями использования](#)
Substack и подтверждаете, что ознакомились с [Уведомлением о](#)
[сборе информации](#) и [Политикой конфиденциальности](#).

У вас уже есть аккаунт? [Войти](#)



данных из
отов — очень

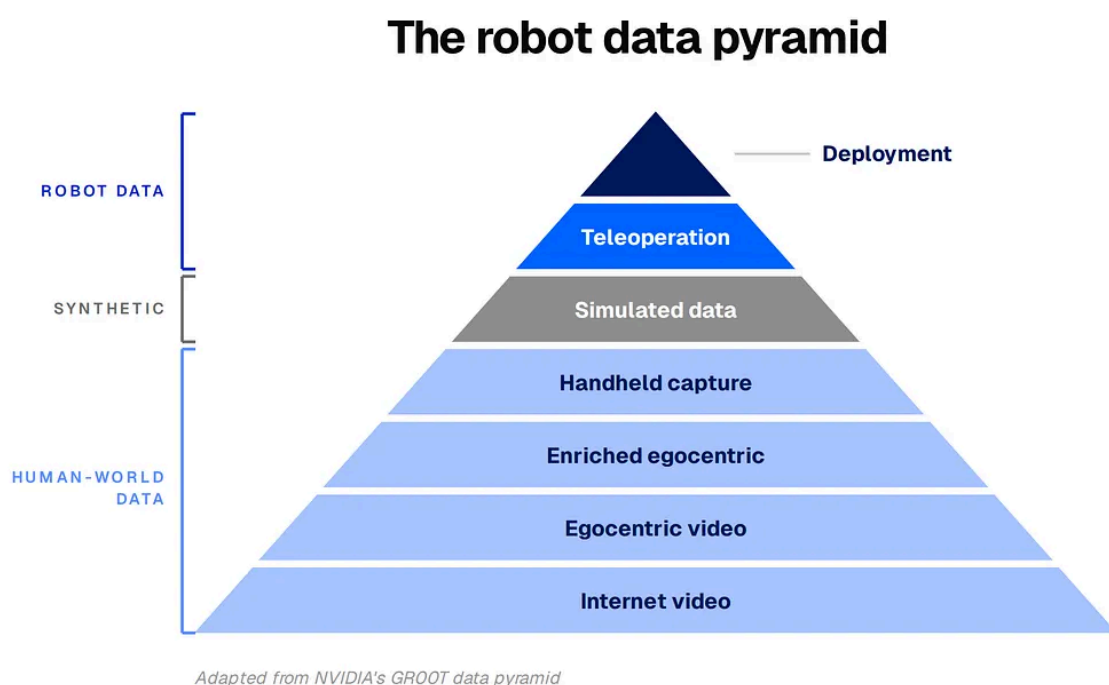
те Dwarkesh
то используе

которые
асколько я по
разница

е от робота, ч
секунду, на
жат гораздо

100 часах данн

Команда GR00T компании NVIDIA предложила рассмотреть идею [пирамиды данных](#): веб-данные в основании, синтетические данные середине, реальные данные о роботах на вершине, с меньшим объемом, но более точными и специфичными для конкретного воплощения. Я разделил пирамиду данных на 7 типов в соответствии с тремя предложенными ими категориями:



Using these different forms allows to address the shortage of robot data, by learning from already available or cheaper to collect data. The lower layers supply priors to models (what objects are, how tasks tend to decompose), and upper layers ground those priors on actual specific hardware and embodiment. Let's talk about each one.

1. Internet video

forces, but can help teach a model about common objects, actions, and environments.

Today it mostly enters robot models indirectly, through the vision-language backbone but various attempts like NVIDIA's [DreamZero](#) and Rhoda's Direct Video Action model are trying to get the most out of this dataset more directly. DreamZero builds its policy on top of a video generation model pre-trained on internet-scale video while [Rhoda AI](#) pre-trains on hundreds of millions of internet videos and then adapting to a given robot with as little as 10 to 20 hours of its data.

2. Egocentric video

Over the last year, many of the robotics labs have increased their purchasing of first-person video from head-mounted cameras. [Egoverse](#) and [Ego4D](#) are open datasets which have 1000s of hours of footage of humans doing common tasks across household and factory / industrial settings.

The interest in this area has come from both approaches like discussed above that pre-train on video, as well as various [research](#) that has found that models pre-trained on carefully filtered egocentric human video beat models pre-trained on the same volume of teleoperated robot data, once both were adapted to the target robot.

3. Enriched egocentric / exocentric

Sometimes, egocentric data can be paired with exocentric data and have additional labels: hand pose tracking, depth, gaze, or even full motion-capture suit. [Ego4D](#) is one example which has around 1,400 hours of first-person and third-person views of skilled tasks.

Tesla for example has also in the past hired workers to perform tasks in motion suits.

tactile forces which are generally one of the biggest gaps with using human

4. Handheld capture

Ultimately, the goal is to collect data in a cheap and easily scalable form that most closely resemble the robot itself. That's the idea of approaches like [UMI](#) (Universal Manipulation Interface), which involves humans going around doing tasks while wearing / holding the end-effectors or grippers that would be on a real robot.

In effect, it allows you to mimic the form factor of the robot without actually needing the robot making it more scalable.

The most notable example leveraging UMI is [Sunday Robotics](#), which was founded by the researchers behind UMI. They ship skill-capture gloves co-designed with their robot's hands into contributors' homes, and say their AC model was trained on around 10 million household episodes collected this way with zero robot data. Another one is Toyota Research Institute's [Large Behavior Models](#) which uses UMI capture alongside teleop and simulation.



Sunday Robotics Data Capture Glove

5. Simulated data

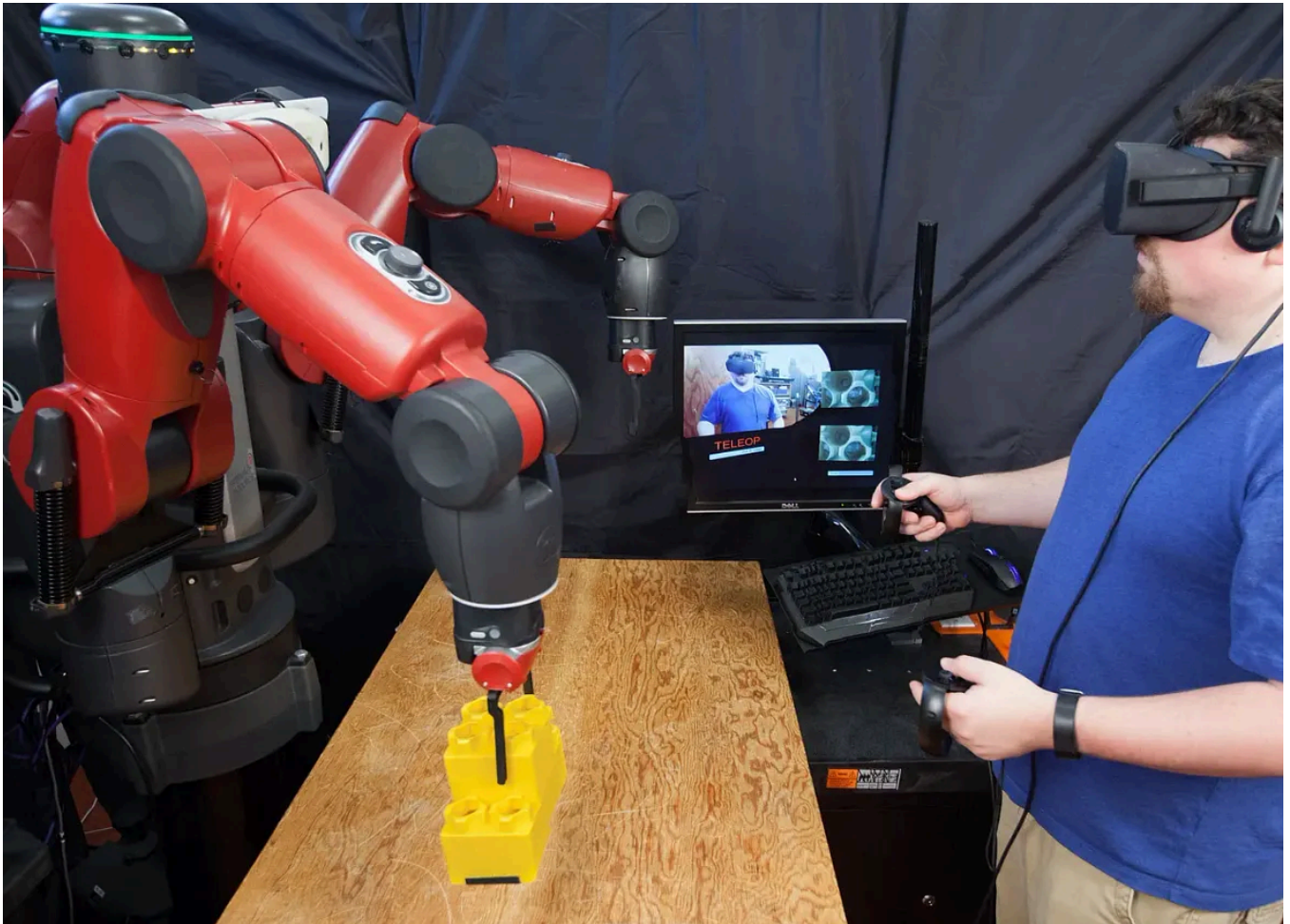
Physics simulators like Isaac and MuJoCo have been around for a while, an increasingly an exciting development is the potential for world models that as learned simulators (NVIDIA's Cosmos, Google's [Genie 3](#)) which could allow robots to evaluate or train their policies in millions of varied virtual environments.

The promise of simulated data is that you're only bound by compute. The downside today is the sim-2-real gap where the results from sim don't transfer perfectly in the real world as well as issues with physics simulators working on manipulation / deformables and world models not being quite ready for prime time yet.

However, World Models like [Waymo's](#) and even [1X's](#) as well as the use of Simulation in the AV industry have shown the promise of this approach. Th

6. Teleoperation

Teleoperation involves someone actively operating the robot, whether virtually through VR setups or through a rig setup where the robot mimics the action



This allows for collecting extremely high-fidelity and essentially real data with an exact embodiment that is needed.

The issue with this approach is the economics and the difficulty to scale. While costs are going down and have been going down, these still cost over \$100 an hour and require relatively skilled operators and you're limited on the number of robots and the setup / environment as well.

Sergey Levine's [Sporks of AGI](#) has a good writeup on the tension around looking for alternative approaches given how difficult it is to scale teleop, but that th

The general view currently amongst labs is that Tele-op will continue to be needed and won't go away, but whether the percent of the training mix can involve a lower percent of tele-op (which could be mostly in post-training) because other sources scale easier.

7. Deployment

Ultimately, the data everyone wants is from real robots doing real work in real settings, ideally work someone is paying for. The robot might run fully autonomously, or with a human teleoperator taking over when it hits trouble similar to how a safety driver handles disengagements in AVs.

The challenge of course is somewhat chicken-and-egg, since you need a robot good enough to deploy before deployment can make it better.

Over time, this data will form the flywheel for companies in terms of deployments leading to data which then helps improve the deployment. Over time, this could become the highest volume data source as deployments increase but today the practical number of deployments is small other than structured environments or in AV.

Two notes on this data source. One, if the deployments are for extremely narrow tasks in extremely structured environments, that data would be helpful to improve the performance of robot policies on those tasks in those environments but may not help generalize to other similar tasks or more varied environments. Two, in the short term, it's likely that most of these deployed robots will be deployed as a hybrid autonomous tele-operated manager where human teleoperators take over when the policy fails. This is both a practical way to get deployments ready in more of these unstructured environments and a really valuable form of data of interventions/failures. The idea goes back to DAGg (dataset aggregation) in imitation learning: run the policy, let a human correct

Closing Thoughts

Progress is coming on all of these fronts at once: approaches to better leverage human video data like Rhoda's and DreamZero, collection being scaled up a egocentric, UMI and teleoperated data, tactile and force capture hardware a collection that is making egocentric data richer in the middle, world models chipping at the sim-to-real gap, and the first real deployment flywheels start to spin somewhat.

No one actually knows how much data is enough. Asked whether robots need 100x or 1,000x more, Sergey Levine's was honest noting: "that's the thing, we know that."

For a robotics team, the more practical question might be, what reliability do we need to get to some level of hybrid deployment given a set of tasks, and what is the cheapest mixture of these sources that clears the reliability bar for that task?

If you're building in physical AI, whether that's robot foundation models, data collection, or the picks and shovels around them, feel free to reach out at tanay@wing.vc.

Subscribe to Tanay's Newsletter

By Tanay Jaipuria · Launched 6 years ago

Musings about tech and business

By subscribing, you agree Substack's [Terms of Use](#), and acknowledge its [Information Collection Notice](#) and [Privacy Policy](#).



[← Previous](#)

Discussion about this post

Comments Restacks



Write a comment...



Jack 2h

How likely is it that Meta makes a comeback in the AI race with wearables? That seems like a good way to capture this kind of data.

♡ LIKE 💬 REPLY



Luke Tremblay 5h

Under the hood, how comparable are “physical AI” systems to modern multi-modal LLMs? We appreciate the comparison as a useful side-by-side for the sake of this discussion on the scaling of training data, I’m genuinely curious how different the model architectures are when processing reasoning through spatial and sensory data compared to tokenized text, audio, or video input. Can the comparison validly extend that far or does the framework for physical AI stand completely apart from the scaffolding that has led to chatbots and code/text-based agentic workflows?

♡ LIKE 💬 REPLY

